DomeSim and The 40 Hour Cabin: the project so far

Written 14 October 2026. Every figure on this page is counted or computed by the project's own code that day.

What it is. One person's open-source project to make a small, owned home cheap enough that the house stops eating the household. It has two halves that keep each other honest: a *method* for framing round buildings from logs split lengthwise with one chainsaw, and the *software* that solves, prices, films and writes about that method, so that no number anywhere is typed by hand. It is built by Donovan Zeanah, a U.S. Navy veteran, in the open at github.com/dkzeanah/DomeSimulator.

The method. A two-frequency geodesic dome, 19.42 ft across with 277 sq ft of floor, framed from 120 wedges split from logs and assembled as 40 flat panels on one jig, joined in pinwheels so nothing meets at a corner; 37.5 hours to frame by the project's clock, about 67 for the whole building. Splitting keeps 89% of a log against 57% sawn into two-by-fours. The product built on it is the *stem cell*: one standard body with a utility core ($1,890), snap-in panels, seams that double as channels for rain, wire and air, and a silicone-soaked cotton skin, listing at $12,036 ($43.45 a sq ft against $217 for a new site-built house), in 15 fit-outs. It stands on a serviced pad: own the building, rent the ground. The same wedges now frame a zome too: 36 panels, every cut one 22.5° bevel, 9 log sections against the dome's 15.

The software. 610 Python files, about 337,364 lines. At its centre is the raw-wedge solver and its 3-D world, which cuts and joins every member; around it, the Dome Creator, the Triangle Infill calculator, the new Wedge Zome world (with rulers, a six-foot person and a click-to-measure readout), cost models for the stem cell, the campaign and a park of pads, the seam-climate controller, the cell wall, the sky-web build, a CNC shop, a book engine whose manuscripts carry live number tokens, a film pipeline that renders narrated lessons in a shared 3-D "Cabin World" in the owner's own synthesised voice and resumes after a crash, a website (zeanahlab.com) that sells the book and runs a dome network, and a recorder that logs how the work gets done.

What has been written. *The 40 Hour Cabin: Complete Edition* (83 chapters in 15 parts, sold on the site with the code); *The 40 Hour Cabin: 2V Geodesic Dome Wedge Method*, the Amazon paperback (55 chapters); the *Master Edition*, a 853-page reader's proof that tells the whole idea in 6 rungs from a bedtime story to published experiments, across 12 threads, with both books bound in; *'Square' on Demand* (32 pages), on two wedges paired into a 3.87 in square only where a square is needed; a title page for *Zome Zoomer*; and *Domology*, a three-in-one book in progress.

What has been filmed. 85 films registered and 34 rendered; 211 video files in deliverables, counting phone cuts and teasers, and 109 PDFs. The newest, *Beyond the Dome*, takes the method past the dome: square on demand, shims, roof pitches, octagons, joints and the zome.

The ask, and who it is for. A grant one-pager and a sheet of efficiency charts ask for $100,650, the sum of costed lines: prove it, then make it repeatable. It is built by a veteran for veterans – 32,882 were homeless on one night in January 2024 (HUD) – to sit beside the programs that already house them.

What is proved, and what is not. Measured: the splitting rate (30 wedges per 4-hour session), strut weights (15-22 kg) and a full-size prototype frame. Modelled: cost, energy, rain, recovery. Not yet: an engineer's rating, an instrumented year of weather, the moulded shell and its tooling. And the numbers that argue back stay on the page – paired squares keep only 53% of a log, less than the mill.

The 40 Hour Cabin: the numbers

What one building costs, what it sells for, what the money buys, and what argues back.

ONE STANDARD BUILDING: COST TO PRICE

MODELLED

Wedge-cut frame	$692
Shower-cap shell	$3,265
Utility column and seal cap	$1,405
Light, fan and cooling	$485
Utility panels	$491
Build labour	$1,884
Materials and labour	$8,221
Overhead	$1,480
Warranty reserve	$329
Cost to build	**$10,030**
Maker's margin, 20% on cost (declared)	$2,006
List price	**$12,036**

THE SAME 277 SQ FT, BOUGHT THREE WAYS

PUBLISHED · MODELLED

	for the floor	a sq ft
New site-built house, without land (NAHB)	$60,107	$217
New manufactured home (Census 2023)	$23,993	$87
This standard building	$12,036	$43.45
... standing on its pad, at cost	$18,276	$65.98

WHAT THE ASK BUYS: $100,650

THE CAMPAIGN'S COSTED LINES

Prove it	$12,850
Make it repeatable	$60,800
Build what an owner cannot	$22,000
Seed the network	$5,000
Margins that repay the ask, at one building's margin each	51 buildings
Startup model: equipment + working capital	$86,000 + $14,000
Materials the working capital buys, at the Budget build's price	2.1 domes

FOR THE OWNER

MODELLED · ILLUSTRATIVE

Heating and cooling, a year: this building / a box of the same floor	$129 / $220
Never spent, every year	$91
Utility core, over its 25-year life	$76 a year
Moving the core to the next building	$260
For comparison: a $100,000 mortgage at 6.5% for 30 years	$227,544 repaid, $127,544 of it interest
This building on its pad, at cost, paid once	$18,276

FOR A PAD HOST: A $10,000 PAD (THE AUTHOR'S TARGET) AT 80% OCCUPANCY

SCENARIO

rent	after costs	payback
$500 a month	$3,880 a year net	30.9 months to repay
$1,000 a month	$8,680 a year net	13.8 months to repay
$1,500 a month	$13,480 a year net	8.9 months to repay
Management, tax and insurance, a year	$920	

CAMPAIGN REWARDS: WHAT EACH TIER LEAVES

MODELLED

tier	pledge	cost to send	kept
The drawings and the numbers	$35	$2	94%
Quilter's kit	$95	$58	39%
One bay's hardware set	$180	$112	38%
A shower cap for your dome	$1,450	$980	32%
The utility hub	$2,400	$1,620	32%
Dome kit, frame included	$19,800	$14,900	25%
Dome kit, bring your own trees	$11,400	$8,100	29%
Pad host's pack	$640	$395	38%

FOR SCALE

PUBLISHED · MODELLED

Unsheltered veterans, one night in January 2024 (HUD)	13,851
A standard building on its pad for each, at cost, before land	$253,146,626

THE NUMBERS THAT ARGUE BACK

COMPUTED

The starter pad, computed, against the $10,000 target	$13,148 ($3,148 over)
Labour in the standard building	about 67 hours, not forty
Two wedges planed square keep of the log / a mill's 2x4s	53% / 57%
Rated by an engineer	not yet -- the first line of the ask
Margin, prices, rents	declared or estimated, each named in the code

Every figure computed by the project's own models on the day this page was printed; modelled figures are the software's, published ones are cited (NAHB, Census, HUD), declared ones are named with their reason in the code. github.com/dkzeanah/DomeSimulator